

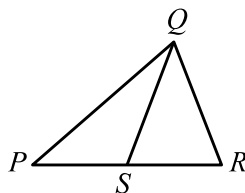
Quantitative Comparison Questions

For Questions 1 to 6, compare Quantity A and Quantity B, using additional information centered above the two quantities if such information is given. Select one of the following four answer choices and fill in the corresponding oval to the right of the question.

- (A) Quantity A is greater.
- (B) Quantity B is greater.
- (C) The two quantities are equal.
- (D) The relationship cannot be determined from the information given.

A symbol that appears more than once in a question has the same meaning throughout the question.

	<u>Quantity A</u>	<u>Quantity B</u>	<u>Correct Answer</u>
Example 1:	$(2)(6)$	$2 + 6$	(A) (B) (C) (D)



	<u>Quantity A</u>	<u>Quantity B</u>	<u>Correct Answer</u>
Example 2:	PS	SR	(A) (B) (C) (D) (since equal lengths cannot be assumed, even though PS and SR appear equal)

The average (arithmetic mean) of 4 donations to a charity was \$80. Two of the 4 donations were \$90 and \$60.

	<u>Quantity A</u>	<u>Quantity B</u>	
1. The average of the other 2 donations		\$80	(A) (B) (C) (D)

AGE DISTRIBUTION OF
EMPLOYEES OF A BUSINESS

Age Interval	Number of Employees
15–24	17
25–34	24
35–44	26
45–54	21
55–64	18
Total	106

Quantity A

Quantity B

2. The range of the ages of the 20 oldest employees of the business

11 years

Ⓐ Ⓑ Ⓒ Ⓓ

Quantity A

Quantity B

3. The sum of the first 7 positive integers

7 times the median of the first 7 positive integers

Ⓐ Ⓑ Ⓒ Ⓓ

Quantity A

Quantity B

4. The number of two-digit positive integers for which the units digit is not equal to the tens digit

80

Ⓐ Ⓑ Ⓒ Ⓓ

In a probability experiment, G and H are independent events. The probability that G will occur is r , and the probability that H will occur is s , where both r and s are greater than 0.

Quantity A

Quantity B

5. The probability that either G will occur or H will occur, but not both

$$r + s - rs$$

(A) (B) (C) (D)

$$S = \{1, 4, 7, 10\}$$

$$T = \{2, 3, 5, 8, 13\}$$

x is a number in set S , and y is a number in set T .

Quantity A

Quantity B

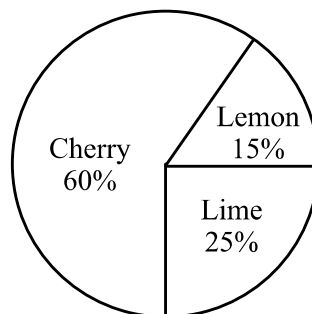
6. The number of different possible values of the product xy

$$20$$

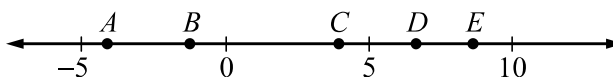
(A) (B) (C) (D)

Multiple-choice Questions—Select One Answer Choice

For Questions 7 to 12, select a single answer choice.



7. The graph above shows the distribution of three different flavors of hard candies—cherry, lemon, and lime—in a candy jar. If all the lemon candies are removed and no other candies are added or removed, what fraction of the remaining candies in the jar will be lime candies?
- Ⓐ $\frac{1}{7}$
- Ⓑ $\frac{2}{9}$
- Ⓒ $\frac{1}{4}$
- Ⓓ $\frac{5}{17}$
- Ⓔ $\frac{5}{12}$
8. R is a list of 15 consecutive integers, and T is a list of 21 consecutive integers. The median of the integers in list R is equal to the least integer in list T . If the two lists are combined into one list of 36 integers, how many different integers are on the combined list?
- Ⓐ 25
- Ⓑ 27
- Ⓒ 28
- Ⓓ 32
- Ⓔ 36



9. From the 5 points A , B , C , D , and E on the number line above, 3 different points are to be randomly selected. What is the probability that the coordinates of the 3 points selected will all be positive?
- Ⓐ $\frac{1}{10}$
- Ⓑ $\frac{1}{5}$
- Ⓒ $\frac{3}{10}$
- Ⓓ $\frac{2}{5}$
- Ⓔ $\frac{3}{5}$
10. In a distribution of 850 different measurements, x centimeters is at the 73rd percentile. If there are 68 measurements in the distribution that are greater than y centimeters but less than x centimeters, then y is approximately at what percentile in the distribution?
- Ⓐ 45th
- Ⓑ 50th
- Ⓒ 55th
- Ⓓ 60th
- Ⓔ 65th
11. Each of the following linear equations defines y as a function of x for all integers x from 1 to 100. For which of the following equations is the standard deviation of the y -values corresponding to all the x -values the greatest?
- Ⓐ $y = \frac{x}{3}$
- Ⓑ $y = \frac{x}{2} + 40$
- Ⓒ $y = x$
- Ⓓ $y = 2x + 50$
- Ⓔ $y = 3x - 20$

12. For a certain distribution, the measurement 12.1 is 1.5 standard deviations below the mean, and the measurement 17.5 is 3.0 standard deviations above the mean. What is the mean of the distribution?
- Ⓐ 13.8
 - Ⓑ 13.9
 - Ⓒ 14.0
 - Ⓓ 14.1
 - Ⓔ 14.2

Multiple-choice Questions—Select One or More Answer Choices

For Question 13, select all the answer choices that apply.

13. Set A has 50 members and set B has 53 members. At least 2 of the members in set A are not in set B . Which of the following could be the number of members in set B that are not in set A ?

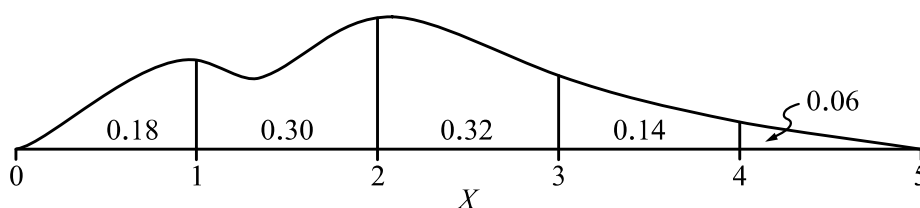
Indicate all such numbers.

- ☐ Ⓐ 3
- ☐ Ⓑ 5
- ☐ Ⓒ 13
- ☐ Ⓓ 25
- ☐ Ⓔ 50
- ☐ Ⓕ 53

Numeric Entry Questions

For Questions 14 to 15, enter your answer in the answer box(es) below the question.

- Your answer may be an integer, a decimal, or a fraction, and it may be negative.
- If a question asks for a fraction, there will be two boxes—one for the numerator and one for the denominator. A decimal point cannot be used in a fraction.
- Equivalent forms of the correct answer, such as 2.5 and 2.50, are all correct. Fractions do not need to be reduced to lowest terms, though you may need to reduce your fraction to fit in the boxes.
- Enter the exact answer unless the question asks you to round your answer.



14. The figure above shows the probability distribution of a continuous random variable X . For each of the five intervals shown, the figure gives the probability that the value of X is in that interval. What is the probability that $1 < X < 4$?

FIVE MOST POPULOUS CITIES IN THE UNITED STATES
APRIL 2000

City	Population (in thousands)
New York	8,008
Los Angeles	3,695
Chicago	2,896
Houston	1,954
Philadelphia	1,518

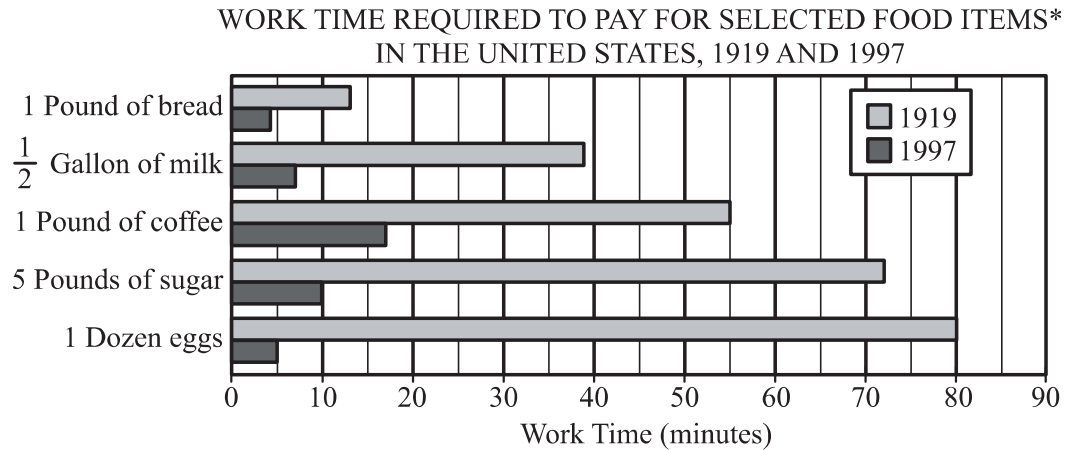
15. The populations of the five most populous cities in the United States in April 2000 are listed in the table above. The total population of the United States in April 2000 was 281,422,000. Based on the data shown, the population of the three most populous cities combined was what percent of the total population of the United States in April 2000?

Give your answer to the nearest whole percent.

 %

Data Interpretation Sets

Questions 16 to 19 are based on the following data. For these questions, select a single answer choice.

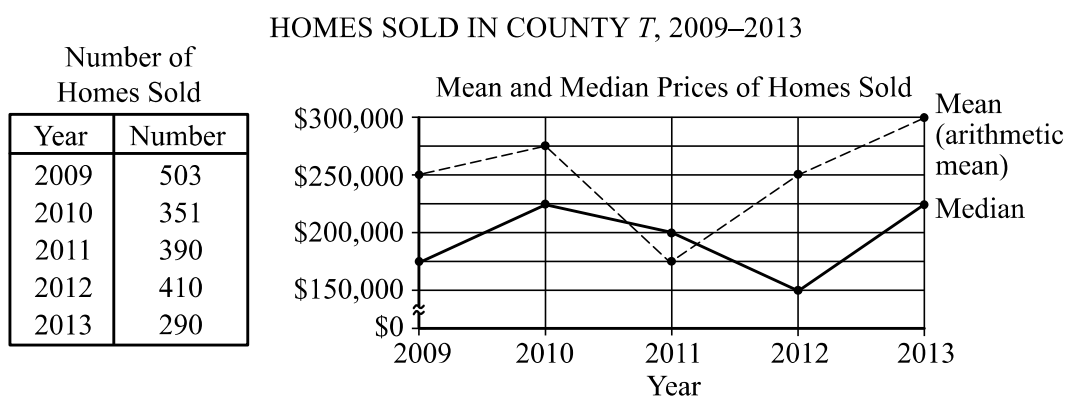


*For each year, the work time, in hours, required to pay for a food item is the average price of the food item divided by the average hourly wage for rank-and-file manufacturing workers. The work time in the graph is given in minutes.

16. In 1997, at the rates shown in the graph, the work time required to pay for which of the following food items was greatest?
- (A) 10 pounds of bread
 - (B) 5 gallons of milk
 - (C) 3 pounds of coffee
 - (D) 20 pounds of sugar
 - (E) 5 dozen eggs
17. If the average hourly wage of the rank-and-file manufacturing worker in 1919 was \$0.55, which of the following is closest to the average price of $\frac{1}{2}$ gallon of milk in 1919?
- (A) \$0.80
 - (B) \$0.65
 - (C) \$0.50
 - (D) \$0.35
 - (E) \$0.20

18. At the rates shown in the graph, which of the following is closest to the number of hours of work time that was required to pay for 20 kilograms of sugar in 1919 ? (1 kilogram equals 2.2 pounds, rounded to the nearest 0.1 pound.)
- (A) 11
(B) 14
(C) 20
(D) 31
(E) 53
19. Eight hours of work time paid for approximately how many more dozen eggs in 1997 than it did in 1919 ?
- (A) 50
(B) 70
(C) 90
(D) 110
(E) 130

Questions 20 to 23 are based on the following data. For these questions, select a single answer choice unless otherwise directed.



20. Which of the following is closest to the mean of the prices of the 700 homes sold in 2012 and 2013 combined?
- (A) \$265,000
(B) \$270,000
(C) \$275,000
(D) \$280,000
(E) \$285,000

21. By approximately what percent did the median price of homes sold in County *T* decrease from 2011 to 2012 ?
- (A) 10%
 - (B) 15%
 - (C) 25%
 - (D) 33%
 - (E) 50%

For Question 22, select all the answer choices that apply.

22. Based on the information given, which of the following statements about the sum of the prices of all the homes sold in a given year must be true?

Indicate all such statements.

- ☐ (A) The sum of the prices for 2010 was greater than the sum for 2009.
 - ☐ (B) The sum of the prices for 2010 was greater than the sum for 2011.
 - ☐ (C) The sum of the prices for 2009 was greater than the sum for 2011.
23. County *T* collected a tax equal to 3 percent of the price of each home sold in the county in 2009. Approximately how much did County *T* collect in taxes from all homes sold in 2009 ?
- (A) \$38,000
 - (B) \$260,000
 - (C) \$380,000
 - (D) \$2,600,000
 - (E) \$3,800,000

Questions 24 to 26 are based on the following data. For these questions, select a single answer choice.

PERSONAL INCOME AND
PUBLIC EDUCATION REVENUE
IN COUNTRY *X*
(in constant 1998 dollars)

Year	Per Capita Income	Revenue per Student
1930	\$6,610	\$710
1940	6,960	950
1950	9,540	1,330
1960	12,780	2,020
1970	17,340	3,440
1980	20,150	4,400
1990	24,230	5,890

24. From 1930 to 1990, approximately what was the average increase per year in per capita income?
- Ⓐ \$150
Ⓑ \$200
Ⓒ \$250
Ⓓ \$300
Ⓔ \$350
25. In 1950 the revenue per student was approximately what percent of the per capita income?
- Ⓐ 8%
Ⓑ 11%
Ⓒ 14%
Ⓓ 17%
Ⓔ 20%
26. For how many of the seven years shown was the revenue per student less than $\frac{1}{5}$ of the per capita income for the year?
- Ⓐ One
Ⓑ Two
Ⓒ Three
Ⓓ Four
Ⓔ Five